# Chapter 21

# Differential Equations

## Only One Option Correct Type Questions

1. The differential equation $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y}$ determines a family of circles with **[IIT-JEE-2007 (Paper-2)]**

(A) Variable radii and a fixed centre at (0, 1)

(B) Variable radii and a fixed centre at (0, –1)

(C) Fixed radius 1 and variable centers along the $x$-axis

(D) Fixed radius 1 and variable centres along the $y$-axis

2. A curve passes through the point $\left(1, \frac{\pi}{6}\right)$. Let the slope of the curve at each point ($x$, $y$) be $\frac{y}{x} + \sec\left(\frac{y}{x}\right), x > 0$. Then the equation of the curve is **[JEE (Adv)-2013 (Paper-1)]**

(A) $\sin\left(\frac{y}{x}\right) = \log x + \frac{1}{2}$

(B) $\operatorname{cosec}\left(\frac{y}{x}\right) = \log x + 2$

(C) $\sec\left(\frac{2y}{x}\right) = \log x + 2$

(D) $\cos\left(\frac{2y}{x}\right) = \log x + \frac{1}{2}$

3. The function $y = f(x)$ is the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^4+2x}{\sqrt{1-x^2}}$ in (–1, 1) satisfying $f(0) = 0$. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)dx$ is **[JEE (Adv)-2014 (Paper-2)]**

(A) $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$

(B) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$

(C) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$

(D) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

4. If $y = y(x)$ satisfies the differential equation $8\sqrt{x}\left(\sqrt{9+\sqrt{x}}\right)dy = \left(\sqrt{4+\sqrt{9+\sqrt{x}}}\right)^{-1} dx,\ x > 0$ and $y(0) = \sqrt{7}$, then $y(256) =$ **[JEE (Adv)-2017 (Paper-2)]**

(A) 80

(B) 9

(C) 16

(D) 3

## One or More Option(s) Correct Type Questions

5. If $y(x)$ satisfies the differential equation $y' - y \tan x = 2x \sec x$ and $y(0) = 0$, then **[IIT-JEE-2012 (Paper-1)]**

(A) $y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{8\sqrt{2}}$

(B) $y'\left(\frac{\pi}{4}\right) = \frac{\pi^2}{18}$

(C) $y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{9}$

(D) $y'\left(\frac{\pi}{3}\right) = \frac{4\pi}{3} + \frac{2\pi^2}{3\sqrt{3}}$

6. Let $y(x)$ be a solution of the differential equation $(1 + e^x)y' + ye^x = 1$. If $y(0) = 2$, then which of the following statement is(are) true? **[JEE (Adv)-2015 (Paper-1)]**

(A) $y(-4) = 0$

(B) $y(-2) = 0$

(C) $y(x)$ has a critical point in the interval $(-1, 0)$

(D) $y(x)$ has no critical point in the interval $(-1, 0)$

7. A solution curve of the differential equation $\left(x^2 + xy + 4x + 2y + 4\right)\frac{dy}{dx} - y^2 = 0$, $x > 0$, passes through the point (1, 3). Then the solution curve **[JEE (Adv)-2016 (Paper-1)]**

(A) Intersects $y = x + 2$ exactly at one point

(B) Intersects $y = x + 2$ exactly at two points

(C) Intersects $y = (x + 2)^2$

(D) Does NOT intersect $y = (x + 3)^2$

8. Let $f : (0, \infty) \to \mathbb{R}$ be a differentiable function such that $f'(x) = 2 - \frac{f(x)}{x}$ for all $x \in (0, \infty)$ and $f(1) \neq 1$. Then

**[JEE (Adv)-2016 (Paper-1)]**

(A) $\lim\limits_{x \to 0^+} f'\left(\frac{1}{x}\right) = 1$

(B) $\lim\limits_{x \to 0^+} xf\left(\frac{1}{x}\right) = 2$

(C) $\lim\limits_{x \to 0^+} x^2 f'(x) = 0$

(D) $|f(x)| \leq 2$ for all $x \in (0, 2)$

9. Let $f : [0, \infty] \to \mathbb{R}$ be a continuous function such that $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t)\,dt$ for all $x \in [0, \infty)$. Then, which of the following statement(s) is (are) TRUE? **[JEE (Adv)-2018 (Paper-1)]**

(A) The curve $y = f(x)$ passes through the point (1, 2)

(B) The curve $y = f(x)$ passes through the point (2, –1)

(C) The area of the region

$\left\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \leq y \leq \sqrt{1 - x^2}\right\}$ is $\frac{\pi - 2}{4}$

(D) The area of the region

$\left\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \leq y \leq \sqrt{1 - x^2}\right\}$ is $\frac{\pi - 1}{4}$

10. Let $\Gamma$ denote a curve $y = y(x)$ which is in the first quadrant and let the point (1, 0) lie on it. Let the tangent of $\Gamma$ at a point $P$ intersect the $y$-axis at $Y_P$. If $PY_P$ has length 1 for each point $P$ on $\Gamma$, then which of the following options is/are correct? **[JEE (Adv)-2019 (Paper-1)]**

(A) $xy' - \sqrt{1 - x^2} = 0$

(B) $xy' + \sqrt{1 - x^2} = 0$

(C) $y = \log_e\left(\frac{1 + \sqrt{1 - x^2}}{x}\right) - \sqrt{1 - x^2}$

(D) $y = -\log_e\left(\frac{1 + \sqrt{1 - x^2}}{x}\right) + \sqrt{1 - x^2}$

11. For any real numbers $\alpha$ and $\beta$, let $y_{\alpha,\beta}(x)$, $x \in \mathbb{R}$, be the solution of the differential equation $\frac{dy}{dx} + \alpha y = xe^{\beta x}, y(1) = 1$ **[JEE (Adv)-2021 (Paper-2)]**

Let $S = \{y_{\alpha,\beta}(x) : \alpha, \beta \in \mathbb{R}\}$. Then which of the following functions belong(s) to the set $S$?

(A) $f(x) = \frac{x^2}{2}e^{-x} + \left(e - \frac{1}{2}\right)e^{-x}$

(B) $f(x) = -\frac{x^2}{2}e^{-x} + \left(e + \frac{1}{2}\right)e^{-x}$

(C) $f(x) = \frac{e^x}{2}\left(x - \frac{1}{2}\right) + \left(e - \frac{e^2}{4}\right)e^{-x}$

(D) $f(x) = \frac{e^x}{2}\left(\frac{1}{2} - x\right) + \left(e + \frac{e^2}{4}\right)e^{-x}$

12. For $x \in \mathbb{R}$, let the function $y(x)$ be the solution of the differential equation **[JEE (Adv)-2022 (Paper-2)]**

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Then, which of the following statements is/are TRUE?

(A) $y(x)$ is an increasing function

(B) $y(x)$ is a decreasing function

(C) There exists a real number b such that the line $y = b$ intersects the curve $y = y(x)$ at infinitely many points

(D) $y(x)$ is a periodic function



## Integer / Numerical Value Type Questions

13. Let $f : R \to R$ be a continuous function which satisfies $f(x) = \int_0^x f(t)dt$. Then the value of $f(\ln 5)$ is

$\frac{dy}{dx}$

**[IIT-JEE-2009 (Paper-2)]**

14. Let $f$ be a real-valued differentiable function on $R$ (the set of all real numbers) such that $f(1) = 1$. If the $y$-intercept of the tangent at any point $P(x, y)$ on the curve $y = f(x)$ is equal to the cube of the abscissa of $P$, then the value of $f(-3)$ is equal to **[IIT-JEE-2010 (Paper-1)]**

15. Let $f : [1, \infty) \to [2, \infty)$ be a differentiable function such that $f(1) = 2$. If $6\int_1^x f(t)\,dt = 3x\,f(x) - x^3$ for all $x \geq 1$, then the value of $f(2)$ is **[IIT-JEE-2011 (Paper-1)]**

16. Let $y'(x) + y(x)g'(x) = g(x)g'(x)$, $y(0) = 0$, $x \in \mathbb{R}$, where $f'(x)$ denotes $\frac{df(x)}{dx}$ and $g(x)$ is a given non-constant differentiable function on $\mathbb{R}$ with $g(0) = g(2) = 0$. Then the value of $y(2)$ is **[IIT-JEE-2011 (Paper-2)]**

17. Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function with $f(0) = 0$. If $y = f(x)$ satisfies the differential equation $\frac{dy}{dx} = (2 + 5y)(5y - 2)$, then the value of $\lim_{x \to -\infty} f(x)$ is ____. **[JEE (Adv)-2018 (Paper-2)]**

18. Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function with $f(0) = 1$ and satisfying the equation $f(x + y) = f(x)f'(y) + f'(x)f(y)$ for all $x$, $y \in \mathbb{R}$. Then, the value of $\log_e(f(4))$ is _____. **[JEE (Adv)-2018 (Paper-2)]**

19. If $y(x)$ is the solution of the differential equation $xdy - (y^2 - 4y)dx = 0$ for $x > 0$, $y(1) = 2$, and the slope of the curve $y = y(x)$ is never zero, then the value of $10y(\sqrt{2})$ is ______. **[JEE (Adv)-2022 (Paper-2)]**

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